Numerical for Practice

Q1. Define declination angle and calculate its value on May 25?

Definition: Declination angle (δ)

The declination angle is the angle between the line joining the centers of the Sun and Earth and the projection of this line on the equatorial plane of the Earth. It represents the angular position of the Sun at solar noon with respect to the Earth's equatorial plane.

- It varies between +23.45° and −23.45° during the year.
- Positive when the Sun is north of the equator (March–September).
- Negative when the Sun is south of the equator (September–March).

Formula: $\delta=23.45 \times \sin[360/365(284+n)]$	(1)

Where n = day number of the year=145

$\delta=23.45 \sin[365/360(284+145)]$

$\delta=23.45\sin[365/360(429)]$

$\delta=23.45 \sin(423.3^0)$

$\sin 423.3^0=\sin 63.3^0 =0.893$, $\delta=23.45\times0.893 = +20.93^0$

The positive sign indicates the Sun is north of the equator.

Q2. a) Calculate the declination angle of beam radiation on a photovoltaic array located in Thane
(25°10′ N, 78°61′ E).

b) What is the value of slope of PV array? c) What is the value of surface azimuth angle?

d) Calculate the latitude angle?

e) Calculate the hour angle?

Given the following data:

Date: October 25, 2007
Time: 10:30 h **(LAT)**
β: 18°
γ: 27°

Answer:

Day number (N) = 298

d) Latitude (φ) = 25°10′ =25.16°

b) Slope of PV array (β) = 18°

c) Surface azimuth angle (γ) = 27°

a) Calculation of Declination Angle (δ)

For October 25 (N = 298):

$\delta = -13.122°$ (using (1))

e) Calculation of Hour Angle (ω)

Local apparent time (LAT) = 10:30 h,

Convert time into decimal hours 10:30 = 10 + 30/60 = 10.5 h

Formula: $\omega = 15$ (12-Solar Time)

$\omega = 15\ (12 - 10.5)$

$\omega = 15\ (1.5)$

$\omega = +22.5°$

3. Calculate the declination angle, latitude angle and hour angle of beam radiation on a plane surface, located at Mumbai at 1:30 PM (LAT) on 15th November. The longitude and latitude of Mumbai are 72° 49' E and 18° 54’ N respectively.

Ans:

Date = **15 November**

Time = **1:30 PM (LAT)**

Longitude of Mumbai = **72°49′E**
Latitude of Mumbai = **18°54′N**

Convert latitude:

$\phi=18^0+54/60=18.9^0$ (Latitude Angle)

n=319

So, $\delta = -19.15^0$ (Declination angle)

Since time is **LAT**, it is already **solar time**.

Soar time =13.5

$\omega=15(12-13.5) = 15(-1.5) = -22.5^0$ (Hour Angle)

4. Calculate the hour angle of beam radiation on a plane surface, located at Mumbai at 1:30 PM **(IST)** on 15th November. The longitude and latitude of Mumbai are 72° 49' E and 18° 54’ N respectively.

Ans

Formula: Solar Time = Standard Time − $4(L_{st} - L_{loc})$ + E

Substitute:

Solar Time=13:30−35.67+14.74

First convert: 13:30=810 min

Solar Time=810−35.67+14.74=789.07 min

Convert to hours: 789.07/60

Solar Time=13.151 h = =13h 9.07 min

ω=15(12-Solar Time) =15(12-13.151) = - 17.27^0

5. Calculate the declination angle, on June 21 and Dec 21 for a surface inclined at an angle of 10 degree and facing due south. The surface is located at Mumbai (19˚07'N, 72˚51'E). Also find surface azimuth angle and latitude angle.

Ans:

β=10˚, φ =19.(07/60)=19.11˚ , γ =0˚ , δ =23.45˚, -23.45˚

6. Determine the LAT corresponding to 14.30 hr (IST) at Mumbai (19°07′ N, 72°51′E) on July 1. In India the standard time is based on 82.50^o E and the time correction on July 1 is (-3.5) min.

Ans:

For India in Eastern hemisphere, it is

$$\text{LAT} = \text{Standard time} - 4\,[L_{st}(\text{in degree}) - L_{loc}(\text{in degree})]\ (\text{min})\ ] + E(\text{min})$$

$$= 14.30\text{hr} - 38.6\text{min} + (-3.5)\ \text{min} = 858\ \text{mins} - 42.1\text{min}$$

$$= 816\ \text{mins} = 13\text{hr} + 36\text{mins} = 13.6\text{hr}$$

7. Calculate I $_{ext}$?

Ans: $I_{ext} = I_{sc}\,[1 + 0.033\cos(360n/365)]\ W/m^2$,

Where, n is the day of the year starting from January 1.

8. Calculate the value of Air Mass if the value of zenith angle is 60^0 ?

Find the value of inclination angle?

Ans: Air mass = $\sec(\theta_z) = 2$ and Inclination angle= $90^0 - 60^0 = 30^0$

9. Solar Thermal Power Plant Cost Calculation (LCC)

The cost of major components of a solar thermal power plant is specified in the following table along with their life spans.

Item	Cost ($/kW)	Life Period (years)
Heat energy collectors	22000	20
Boiler + steam turbine	11900	10
Electric generator	6500	10
Accessories and tools	500	5

If the rate of interest is 15% and the maintenance cost is 2% of the capital cost per year, calculate:

(a) Capital cost per kW
(b) Maintenance cost per kW
(c) Life cycle cost (LCC) per kW

Ans: (a) Capital Cost per kW

Total capital cost = 22000 + 11900 + 6500 + 500

Total capital cost = 40900 $/kW

Answer: Capital cost per kW = 40900 $/kW

(b) Maintenance Cost per kW
Maintenance cost per year = 2% of capital cost

Maintenance cost = 0.02 × 40900
Maintenance cost = 818 $/year

Using Present Worth Factor for i = 15% and n = 20 years:

Present worth factor , $P=A[((1+i)^n-1)/(i(1+i)^n)] = 10.92$

Where A=818 , i=0.15, n=20
Maintenance cost (present worth) = 818 × 10.92
Maintenance cost = 8929.47 $/kW

Answer: Maintenance cost per kW = 8929.47 $/kW

(c) Life Cycle Cost (LCC)
LCC = Capital cost + Maintenance cost

LCC = 40900 + 8929.47
LCC = 49829.47 $/kW

Answer: Life cycle cost (LCC) per kW = 49829.47 $/kW

10. (a) Calculate the declination angle of beam radiation on a photovoltaic array located in Thane
(25°10′ N, 78°61′ E). Given the following data:

Date : October 25, 2007
Time : 10:30 h (LAT)
Slope of PV array (β) : 18°
Surface azimuthal angle (γ) : 27°

Also calculate:(b) Latitude angle (c) Hour angle

Ans:

Day number for October 25: $n = 298$

1. Declination Angle (δ)
Formula:
$\delta = 23.45 \sin [360/365 (284 + n)]$

Substitute $n = 298$

$\delta = 23.45 \sin [360/365 (284 + 298)]$
$\delta = -13.122°$

Declination angle = $-13.122°$

2. Latitude Angle (φ)
Latitude given = 25°10′ N

Convert to decimal:
$\varphi = 25 + (10/60)$
$\varphi = 25.16°$

Latitude angle = 25.16°

3. Hour Angle (ω)
Time given = 10:30 (LAT)

Convert to decimal:
$10 + 30/60 = 10.5$ hours

Formula:
$\omega = 15 (12 - \text{Solar time})$

$\omega = 15 (12-10.5)$
$\omega = 15 (1.5)$
$\omega = + 22.5°$